

Student Result Card System Report

1. Introduction

The Student Result Card System is a simple console-based application developed in C++. The system is designed to manage student academic records and generate result cards automatically.

2. Objectives

- To store student academic information.
- To calculate total marks and percentage.
- To assign grades automatically.
- To generate a student result card.
- To validate student name and roll number.
- To predict future grades based on expected marks.
- To provide a menu-driven system for user interaction.

3. Problem Statement

Managing student results manually can be time-consuming and may lead to calculation errors. Educational institutions require a simple and reliable system for storing student records, calculating results automatically, and displaying grades accurately.

4. Functional Requirements

1. Enter student information
2. Validate student name using alphabets only
3. Validate roll number using numbers only
4. Input marks for different subjects
5. Calculate total marks, percentage, and grade
6. Display complete result card
7. Predict future grades
8. Provide a menu-driven interface

5. Non-Functional Requirements

Usability: The system should be easy to use through a simple console interface.

Reliability: The calculations should produce accurate results.

Maintainability: The code should be simple and easy to modify.

Performance: The system should respond quickly to user input.

6. Tools and Technologies Used

Tool	Purpose
C++	Programming Language
iostream	Input and Output Operations
Visual Studio 2022	Development Environment

7. System Workflow

9. Display menu options
10. User selects an option
11. Enter student information
12. Validate student name and roll number
13. Enter subject marks
14. Calculate total, percentage, and grade
15. Display result card
16. Predict grade if selected
17. Exit program

8. Grade Criteria

Percentage	Grade
90 and above	A
80 – 89	B
70 – 79	C
60 – 69	D
Below 60	F

9. Conclusion

The Student Result Card System is a simple and effective C++ project that automates the process of result calculation and result card generation. It demonstrates the use of loops, conditions, validation, and menu-driven programming concepts.